

Letter of Intent: FlowStun

Inline electrical stunning and insensibility verification for small pelagic fish on vessels

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Requested funding: USD 410,300 over 12 months

Online: [Interactive proposal site](#) · [3D CAD model in kernelCAD](#)

Summary

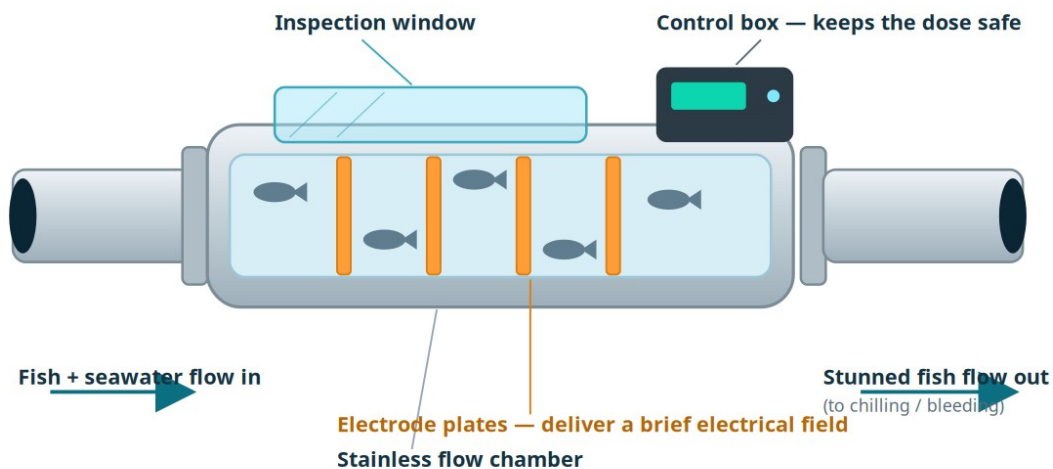


Figure 1 — FlowStun inline stunning module: fish flow left to right through a wet chamber with a glass inspection window, electrode cartridge (orange), and control enclosure.

Most small wild fish are killed while still conscious.

Anchovy, sardine, sprat, herring and menhaden are pumped, crowded, brailled, and then chilled in slurry or sent straight to bleeding, all without stunning. The gear is built to move tonnes per hour rather than to handle animals individually. Salmon are the rare exception, and they account for under 1% of farmed fish. The numbers everywhere else are enormous: a single one-tonne-per-hour lane running 10–20 g fish handles 50,000–100,000 animals an hour.

FlowStun is a prototype aimed to realistically fit into the process: the transfer point where fish leave the pump or brail and enter the hold, the ice slurry, or the bleed line. It is a short nonconductive wet channel with replaceable electrode cartridges.

Fish pass through in the existing flow of seawater, and the channel delivers a controlled electric field while measuring what reaches them - voltage, current, water conductivity, temperature, flow speed, and how long each fish spends in the field.

The firmware regulates the waveform against the measured load, so a shift in salinity, a fouled electrode, or a change in fish density is detected instead of quietly altering the dose. Every run is recorded.

This is not an attempt to show that electricity can stun fish; that has been demonstrated for decades in trout and salmon. It is a cheap, auditable module which can deliver a *reliable* stun in dense, moving, conductive seawater on a working vessel, and whether we can prove the fish were genuinely unconscious.

That proof is the central difficulty, and the project is built around it. Stopping movement is not the same as ending awareness.

A fish that is paralysed but conscious, or that recovers before the downstream process kills it, is a worse outcome than doing nothing, because it looks like success.

So FlowStun keeps two things apart:

- biological validation, done by a contracted fish-welfare laboratory that defines and measures unconsciousness;
- field verification, which only claims that exposure stayed inside an envelope the lab has validated, backed by sample reflex checks and audit logs.

Reversible immobilisation counts as a failure in this project.

In 12 months I will deliver a bench-tested prototype, a non-live engineering dataset, a validated - or disproven - humane exposure envelope for one small-pelagic species, a vessel-realistic pilot design with costs, and a clear go/no-go recommendation.

The problem and who it affects

Coefficient's RFP asks for stunning methods that are immediate, lasting, verifiable, and practical under real fisheries and aquaculture constraints, with explicit attention to vessels short on power and space and to small wild fish caught in bulk. FlowStun targets that case.

Small pelagics are a different problem from salmon farming. They arrive in dense, uneven surges. They get pumped through pipes, crowded, tipped into tanks, and cooled while alive. Throughput is the whole point of the operation.

Any welfare step that needs fish oriented one by one, slows unloading, eats deck space, leaves residues, or demands constant crew attention **will not be adopted**, no matter how humane it looks on paper.

What makes the segment worth the effort is its sheer individual-animal scale.

At 1,000 operating hours a year, one lane touches roughly 50–100 million fish. A modest fleet or a few processing points running parallel lanes reaches hundreds of millions to over a billion fish a year. That arithmetic is the reason a low-cost inline method is worth testing.

The existing humane methods do not transfer here. Percussive stunning works when fish can be positioned individually, but small pelagics move as bulk flow.

Ice slurry and asphyxia do not produce immediate, dependable insensibility.

Gas and chemical routes bring residue, food-safety, crew-safety, and regulatory problems.

Electrical stunning is the physically plausible candidate, yet most existing systems were built for controlled farm flows, larger animals, known water chemistry, or fixed plant infrastructure.

The open question is whether a safe, affordable, marine-grade module can deliver a uniform enough dose in conductive water packed with fish.

How it works

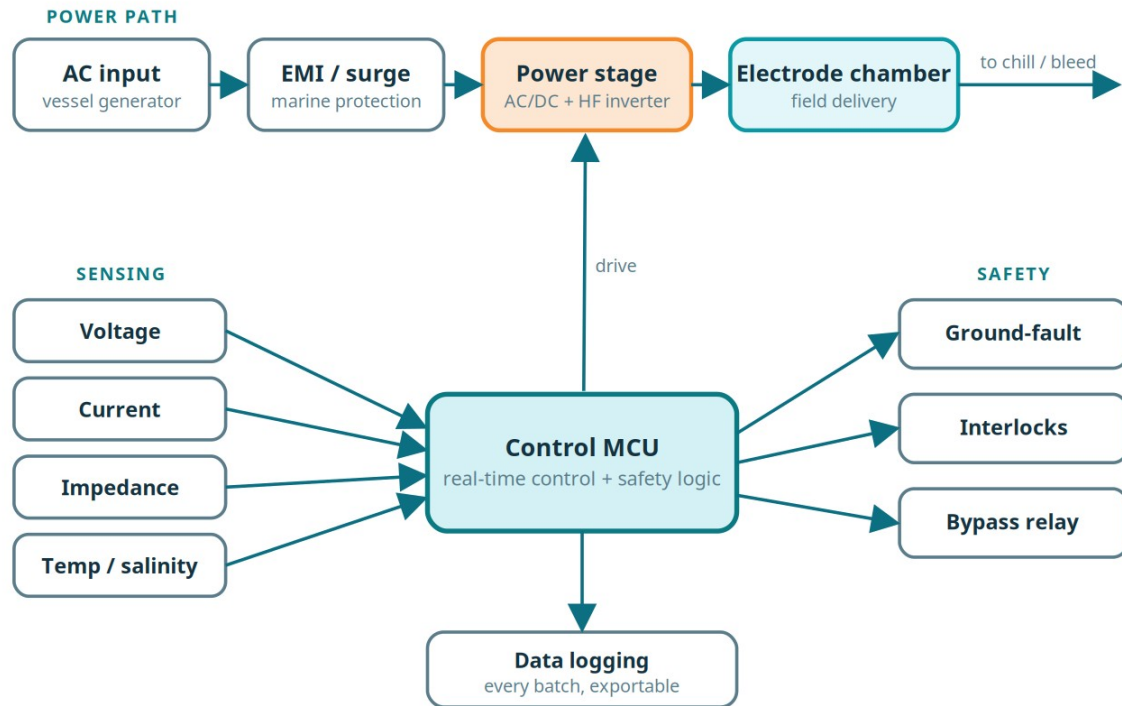


Figure 2 — Power and control architecture: the controller regulates delivered dose against measured load and trips to a fail-safe bypass on any out-of-envelope, safety, or flow fault.

The prototype has five parts.

The exposure chamber is a food-contact-safe nonconductive channel with electrode cartridges that swap out, so we can compare geometries directly: parallel plates, segmented rings, opposed curved plates, and shielded interdigitated layouts.

The variables I will map are channel length and cross-section, electrode spacing, baffling, residence-time spread, pressure drop, field uniformity, and how hard the thing is to clean. The commercial intent is several small parallel lanes rather than one large high-current chamber. Parallel lanes mean safer currents, simpler maintenance, and an easy way to scale to different vessel sizes.

The power stage runs off isolated ship or shore power with current limiting and monitored output, driving an AC or biphasic pulsed waveform. The first engineering envelope covers roughly 100–600 V/m equivalent field strength, 0.5–3.0 s dwell, salinity from brackish to full seawater, and power within small-vessel limits. Those are design ranges for the hardware, not welfare claims. The controller

closes the loop on measured current, voltage, conductivity, and impedance, so fouling, air gaps, and density swings show up as data rather than as silent dose changes.

The sensor layer measures the conditions that set the real exposure: voltage, current, sampled load response, conductivity, temperature, flow, chamber fill, a turbidity or density proxy, and residence time.

Fault logic flags air entrainment, shorts, fouled electrodes, overheating, overload, density outside the validated range, and bypass operation. Safety hardware is not optional on a wet, high-current device near crew. It includes galvanic isolation, ground-fault detection, emergency stop, cover interlocks, output discharge paths, current and temperature limits, sealed marine connectors, and a fail-open mechanical bypass so a disabled prototype never blocks the vessel's own process.

Biological instrumentation links the engineering variables to welfare endpoints, but only as supporting evidence. High-speed video can measure motion, posture, convulsion, and recovery, yet none of that proves unconsciousness. The welfare partner sets the primary endpoints and signs off on them before any trial. If electrophysiology is not practical at anchovy body size, the protocol will say so and justify a behavioural and reflex battery plus a recovery-monitoring window instead. Endpoints are fixed before we see results, not after.

The data layer is what makes the output evidence. Each run logs the waveform, electrode geometry, chamber configuration, calibration state, water properties, flow, density proxy, exposure time, endpoint scores, recovery notes, adverse events, and operator comments. The deliverable includes the structured tables, firmware logs, interface documentation, and a cost and adoption model the reviewers can compare against other approaches.

Plan and timeline

Months 1-3; engineering, no live animals. I build the first chamber, the cartridges, the isolated power stage, the sensor board, the firmware, and the logging software, then characterise them against seawater and brine, resistor and gel loads, any carcass or unavoidable-mortality samples available, and controlled flow loops. The goal is to understand field uniformity, load behaviour, heating, residence-time spread, clogging, cleaning effort, pressure drop, and failure modes before any fish are involved. I will count this phase a success if the output stays

regulated across a realistic range of conductivity and load, faults are detected and trigger safe shutdown, and two or three electrode geometries are ready for biological testing. The model species is locked here, in Month 1: European anchovy (*Engraulis encrasicolus*) first choice, European sardine (*Sardina pilchardus*) as backup if the welfare lab finds supply or handling more reliable.

Months 4–6; partner protocol and a smaller search space. The contracted fish-welfare lab finalises species, endpoints, sample sizes, stopping rules, ethics approval, and the downstream kill method. Non-live tests with the chosen species look at blockage, mechanical damage, conductivity effects, and residence-time variation. The point of this phase is to cut the live-trial matrix down to a few pre-specified envelopes. If the partner cannot secure the species, protocol, or approval on a believable schedule, the project stops here and returns the engineering findings rather than running weak biology.

Months 7–9; live validation. Live work happens only in an approved facility under the partner's protocol, comparing the pre-specified envelopes against sham or current-practice controls where ethics allow. The outcomes are immediate insensibility, no recovery before death under the paired downstream process, and no unacceptable handling harm. Alongside the engineering data: dose stability, power draw, fault rate, density tolerance, sensitivity to water chemistry. If no humane envelope appears within safe, practical limits, the project ends before any vessel deployment.

Months 10–12; pilot design and adoption package. I harden the design for washdown, corrosion, vibration, connector abuse, cartridge replacement, cleaning, bypass, mounting, and operator workflow. The pilot concept is a 1–2 tonne/hour lane, parallelised for higher throughput, drawing single-digit kW per lane if the validation supports it, with quick-swap wet parts and a simple status output. I will pursue a dry or non-live dockside test with a processor or vessel contact. The final outputs are the pilot design, cost model, risk register, and a full-proposal plan if the welfare data justify going on.

Budget

I have requested **USD 410,300**, weighted toward the science that decides the outcome rather than toward my own time.

Item	USD
Fish-welfare laboratory subcontract — protocol, endpoints, ethics, live trials	154,000
Engineering lead (me), full-time 12 months — firmware, control, power, integration, data	92,000
Prototype build — chamber, cartridges, enclosure, pumps, plumbing, sensors, safety hardware	46,500
Power electronics, isolation, DAQ, rented high-speed camera, calibration	24,800
Compliance, animal-use administration, insurance, legal	26,000
Mechanical and marine fabrication support (contract)	23,500
Animal sourcing, shipping, partner-site logistics, travel	16,200
Contingency	27,300

A few choices are deliberate. The welfare subcontract is the largest line because it is the part that determines whether FlowStun works as a welfare device at all. My own labour is set at a fully-loaded \$92k for twelve months of full-time embedded, power, and control work, below typical Western engineering contract rates, and well under the subcontract, because a single founder's time should not be the most expensive thing a welfare funder pays for. Compliance and contingency are funded as real lines rather than rounding, because live vertebrate trials across jurisdictions carry administrative and schedule risk, and the project is structured to be stopped early if the biology does not hold.

Who does the work

I lead engineering as a Founder: requirements, firmware, control architecture, power integration, data logging, test automation, prototype assembly, and risk management. My background is embedded firmware and hardware-facing systems. Through LabWired I build deterministic firmware simulation and virtual-hardware validation for embedded teams - microcontroller targets, CI workflows, hardware-in-the-loop setups, structured logging, reproducible test runs. That is directly relevant to the parts of FlowStun: deterministic control, sensor calibration, fault handling, electrical-safety assumptions, and audit logs.

The project needs two outside capabilities before any live work. The first is a fish-welfare scientist or aquatic veterinarian who owns the biological validation and the animal-use protocol. This is a named critical-path dependency, and I am securing a partner ahead of any award execution. The second is a mechanical or marine fabricator familiar with wet food-processing or vessel equipment. I will also recruit a processor or vessel operator for workflow review and a non-live dockside test. I will not put forward a field-ready welfare claim without biological sign-off.

Securing the welfare-science partner and at least one vessel or processor contact is the first action on award, and I am beginning that outreach now rather than after funding — the proposal's credibility depends on naming them before any live work.

Adoption, cost, and regulation

The device has to fit an existing transfer point with no individual handling, no major retraining, no chemicals, no extra deck area, and no slower unloading. What the operator sees is an inline wet module with self-test, bypass, a status light, swappable electrode cartridges, and downloadable logs. Maintenance should feel like cleaning and replacing wet parts, not running a lab instrument. Hardware cost target is \$8,000–20,000 per lane excluding installation, with running cost driven mainly by electricity and electrode wear. If one lane handles 50–100 million fish a year, the cost per fish can be low enough for thin-margin pelagic economics, but only if the throughput and cleaning numbers turn out real instead of assumed, which is part of what the prototype is for.

On regulation, I will not claim clearance inside 12 months. Rules differ across wild capture, aquaculture, and reduction, and across jurisdictions. The chamber uses food-contact-safe, cleanable materials with no residues and no consumable stunning agents, with documented electrical-safety controls. What this project

produces is a map of the likely approval pathway, the vessel-safety and electrical-compliance documentation needed, and any obvious blockers, before anyone proposes a fleet pilot.

Risks and the go/no-go decision

The biggest welfare risk is mistaking stillness for unconsciousness - a fish that has stopped moving but can still feel. Four safeguards: the welfare lab (not the device) defines and independently proves what unconsciousness is, so the unit never "declares" a fish stunned from its own sensors; every trial includes a recovery-watch window to catch fish that wake before they die; the operating settings run deliberately stronger than the proven minimum; and a random sample of fish is reflex-checked in the field against the lab's endpoints.

The engineering risk is a non-uniform field in conductive water packed with fish. I manage it by mapping the chamber, swapping modular geometries, measuring residence time, capping density, and using parallel lanes instead of one big chamber.

The adoption risk is a device that clogs, slows transfer, shocks crew, corrodes, or is a pain to clean. Marine-grade isolation, a reliable bypass, washdown design, quick-swap cartridges, and early operator review are how I keep that in check.

The scale risk is that results hold for one species and no further. I manage it by choosing a high-impact model, stating the limits plainly, and only extending to the next species if the first data are strong.

A **go** decision needs all of:

immediate insensibility at realistic density and flow; no recovery before death under the paired downstream process;

stable electrical control across realistic water conditions; no unacceptable tissue or handling harm; a credible path to at least one tonne/hour per lane; and a mature hardware cost that fits low-margin fisheries.

A **no-go** follows if insensibility cannot be told apart from immobilisation, if recovery happens before death, if the field cannot be made uniform in dense conductive flow, if power or safety needs are incompatible with vessels, or if operators would have to accept process disruption they never would in practice.

Why fund this

FlowStun is a disciplined test of a high-upside case the RFP calls out: small wild fish in huge numbers, on vessels and lines where space, power, verification, and economics block conventional humane slaughter.

It does not assume electrical stunning is automatically humane. It asks whether controlled field geometry, closed-loop power, and real biological validation can turn a known physical effect into a practical pre-kill welfare step for one of the largest individual-animal segments in food production.

A positive result justifies a larger fleet or processor pilot and could eventually reach a billion or more fish a year. A negative result is still worth the money: it would show whether seawater conductivity, dense flow, recovery, power, handling, or cost is the thing that rules this route out, and save the next funder from finding out the expensive way.

Either way, what Coefficient gets back is evidence to decide on.

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